

Improved extraction of carotenoids from orange peel using pH-switchable natural deep eutectic solvents: preparation and stability analysis of nanoemulsions

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ABSTRACT

To promote the valorization of orange peel, carotenoids were extracted from orange peel powder using a pH-switchable hydrophobic natural deep eutectic solvent composed of decanoic acid and lauric acid (2:1, mol/mol). Extraction conditions were optimized as 40.50 °C, 10.5:1 mL/g, and 16.80 min, giving a carotenoid yield of 60.98 ± 0.87 mg/100 g. The extract exhibited high antioxidant activity, with DPPH and ABTS radical scavenging rates of 88.97% and 94.85%, respectively. An oil-in-water nanoemulsion prepared from the carotenoid-enriched extract showed a mean droplet size of 76.78 ± 1.58 nm, a polydispersity index of 0.15 ± 0.01, and a zeta potential of -42.56 ± 1.59 mV. In contrast to conventional solvent-based extraction, this work integrates green extraction and nanoemulsion delivery using a pH-switchable solvent system. The nanoemulsion exhibited good stability under different environmental conditions, highlighting its potential for food applications.

1. Introduction

Orange peel is a significant byproduct of the citrus industry, generating over 10 million tons of waste annually. While a small portion is used as animal feed additives, the majority is incinerated or landfilled. This practice undermines sustainable development, causing environmental pollution and resource wastage (Murador et al., 2019). Orange peels are rich in bioactive compounds, including carotenoids, polyphenols, pectin dietary fiber, and minerals (Xu et al., 2019). Based on their chemical structure, carotenoids are primarily divided into two categories: (i) carotenoids-linear or cyclic hydrocarbon chains (e.g., α -carotene, β -carotene, lycopene); and (ii) xanthophylls-oxygen-containing derivatives of carotenoids (e.g., zeaxanthin, lutein, β -cryptoxanthin (Vilas-Boas et al., 2025). Carotenoids are widely used lipophilic pigments among natural pigments, with more than 750 types discovered in nature to date (Desmarchelier & Borel, 2017). In addition to their excellent coloring properties, carotenoids also exhibit antioxidant, anti-aging, and anticancer activities (Boonlao et al., 2022). As a readily available and low-cost resource, the efficient recovery of carotenoids

from orange peel can effectively address environmental pollution issues while maximizing their potential in the food industry.

Deep eutectic solvents (DESs) are a novel class of green solvents proposed by Abbott et al. (2004). They consist of binary or ternary eutectic mixtures composed by hydrogen bond acceptors (HBAs) and hydrogen bond donors (HBDs) in a combination of particular stoichiometry. The solidification points of these blends are significantly lower than the melting points of their neat components. Compared to traditional organic solvents and ionic liquids, DES offer multiple advantages that have led to their widespread adoption, including low volatility, low flammability, biodegradability, and high biocompatibility. As a valuable tool for realizing the concept of green chemistry, DES can be classified into two categories based on their water solubility: hydrophilic DES and hydrophobic DES (Ma et al., 2019). Abbott first synthesized hydrophilic DES at room temperature using choline chloride and urea, and verified the formation of hydrogen bonds during the synthesis via NMR hydrogen spectroscopy characterization. Given the hydrophilic nature of many biologically functional molecules, most research has focused on the application of hydrophilic DES. However, hydrophilic DES face

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significant challenges in effectively extracting hydrophobic compounds found in natural products, such as carotenoids and tocopherols (Gao et al., 2020). In 2015, the Van Osch et al. (2015) team first synthesized and proposed hydrophobic eutectic solvents using long-chain carboxylic acids and quaternary ammonium salts as raw materials. As research into hydrophobic DESs deepened, the concept of reversible or switchable phase transitions in DESs emerged, leading to the development of switchable deep eutectic solvents (SDES). Under specific external conditions, SDES can undergo reversible transitions between water-soluble and water-insoluble states, enabling rapid, efficient, and sustainable extraction and recovery of target compounds. Longeras et al. (2020) triggered the phase transition of a DES composed of lidocaine and oleic acid by altering the temperature of the system. The emergence of SDES has provided new opportunities for greener extraction processes. Their reversible switching behavior may facilitate downstream separation and solvent recovery, while the low volatility and tunable properties of DES-based systems may help reduce the use of conventional organic solvents and enhance process sustainability (Jessop et al., 2012; Paiva et al., 2014).

In recent years, the food industry has been exploring the incorporation of various bioactive substances—such as polyphenols, vitamins, and lipids—into food matrices. However, these substances exhibit low bioavailability and poor stability in their free states. To overcome these challenges, nanotechnology has gained widespread application in both food and pharmaceutical fields by encapsulating and protecting these bioactive substances at specific rates, thereby ensuring their targeted delivery. Zhou et al. (2025) prepared a camellia red A-lysozyme (CA-LYS) nanoemulsion using camellia red A (CA) and lysozyme (LYS). The resulting nanoemulsion effectively protected lutein from ultraviolet irradiation, inhibited lipid oxidation, and exhibited strong stability under acidic, neutral, and high-temperature conditions. The inherent properties of carotenoids, such as poor water solubility and susceptibility to oxidation, hinder their absorption and utilization in the human body. Particularly when exposed to light, high temperatures, and oxygen, the structural stability of carotenoids is severely compromised, leading to a significant reduction in their biological and functional activity. This limitation has, to some extent, impeded their widespread application in food, pharmaceuticals, and nutritional supplements (Li, Li, et al., 2020). The preparation of nanoemulsions can help slow down carotenoid degradation, effectively addressing the aforementioned issues. Bu et al. (2023) utilized glycosylated soy protein isolate to construct nanoemulsions for β -carotene loading. The nanoemulsions exhibited an average particle size below 500 nm, with β -carotene retention exceeding 80% during 28 days of storage. Following in vitro digestion, the bioavailability of β -carotene in the emulsion increased to over 70%. The results indicate that saccharified soy protein nanoemulsions exhibit excellent storage stability and are suitable for encapsulating and delivering β -carotene.

This study proposes an integrated strategy for the extraction, recovery, and utilization of carotenoids from orange peel using a switchable hydrophobic DES system. A hydrophobic DES was first used to extract carotenoids. The novelty of this work lies in the subsequent solid-state recovery process, which simplifies the overall workflow, reduces the number of separation steps, and provides a greener alternative to conventional solvent-based recovery methods. After the DES extract was mixed with water, polyetheramine D-230 was added to induce in situ solvent switching, transforming the biphasic system into a homogeneous hydrophilic phase. This switch altered the solvation environment and decreased carotenoid solubility, leading to carotenoid precipitation and enabling their recovery by centrifugation. This approach minimizes the use of hazardous organic solvents and improves the environmental friendliness and energy efficiency of the process. Furthermore, the DES extract was directly used as the oil phase to prepare a carotenoid-loaded O/W nanoemulsion. The effects of physicochemical factors on nanoemulsion stability were evaluated to enhance the dispersibility and stability of carotenoids in aqueous environments. Overall, this study

demonstrates the potential of integrating extraction, recovery, and formulation as a route for the value-added utilization of orange peel-derived carotenoids.

2. Materials and methods

2.1. Reagents and materials

Decanoic acid (97%) and octanoic acid (98%) were purchased from Shanghai Xianding Biotechnology Co., Ltd. Lauric acid (98%), nonanoic acid (96%), and polyetheramine D-230 were obtained from Shanghai McLean Biochemical Technology Co., Ltd. (–)-Menthol (99.5%) and Tween 80 were purchased from Aladdin Reagent Co., Ltd. β -carotene standard, 1,1-diphenyl-2-hydrazinylhydrazine, and 2,2'-bis (3-ethyl-benzothiazol-6-yl) sulfonic acid were purchased from Shanghai Yuanye Bio-Technology Co., Ltd. All other chemicals and reagents employed in this research were of analytical pure grade.

The oranges used in this experiment were sourced from Hubei Province, China. Only the orange peels were used for subsequent experiments. Fresh oranges were washed, peeled, and the peels were cut into approximately 1 cm $\times$ 1 cm cubes. These cubes were pre-frozen at -80°C for 24 h, followed by freeze-drying for 48 h. The freeze-dried samples were pulverized using a tissue grinder, filtered through an 80-mesh sieve, and stored at room temperature in a dry, dark environment until use.

2.2. Preparation of DESs

The preparation of DES followed the method of Dai et al. (2015), namely the heating method. Decanoic acid and lauric acid were mixed at a molar ratio of 2:1 and stirred magnetically at 70°C for 2 h until a uniform, transparent liquid completely free of any precipitate formed. Prior to use, the DES was left at room temperature overnight without recrystallization. The DES screening method is detailed in the supplementary materials (Table S1 and Fig. S1). Based on the DES screening results (Fig. S1), decanoic acid:lauric acid (2:1) was selected for subsequent experiments considering not only its extraction performance but also its suitability for the following solvent-switching recovery and nanoemulsion formulation. Since the molar ratio of DES constituents may significantly influence solvent polarity, viscosity, intermolecular interactions, and mass transfer behavior, further investigation of different component ratios would be valuable for improving extraction efficiency and clarifying DES design principles. In addition, the controlled incorporation of small amounts of water may alter solvent structure and transport properties, thereby affecting extraction behavior. Similar compositional effects and solvent-design strategies have been discussed in recent studies on natural deep eutectic solvent systems for anthocyanin extraction and stabilization (Huang et al., 2024; Huang, Wang, et al., 2025). Comparative evaluation with other hydrophobic DES systems would also be helpful for further contextualizing the relative advantages and limitations of DES.

2.3. Extraction of carotenoids

Precisely 1 g of orange peel powder was weighed into a 50 mL centrifuge tube. Then, 10 mL of a deep eutectic solvent was added to the tube, and the mixture was heated at 50°C for 20 min under magnetic stirring. Subsequently, the mixture was placed in a centrifuge and spun at 8000 rpm for 10 min (TG16G, Shanghai, China). Finally, the DES extract was collected and stored at 4°C in the dark for no longer than 24 h before use.

2.4. Establishment of a carotenoid detection method

A specified amount of β -carotene standard was prepared in anhydrous ethanol, and the UV–visible spectrophotometer (UV2500PC,

Japan) was used to scan the entire spectrum from 300 to 600 nm to determine the maximum absorption wavelength $\lambda_{\max}$ of β -carotene.

The standard β -carotene sample was enumerated at 1 mg and solved in 10 mL of anhydrous ethanol. The stock solution was subsequently diluted to formulate a number of standard solutions with respective concentrations of 1, 2, 4, 6, 8, 10, and 12 $\mu\text{g/mL}$. The absorbance of these solutions was calculated to establish a calibration curve utilizing a UV-visible spectrophotometer.

2.5. Analysis of carotenoid content in orange peel

Quantitative analysis was performed by measuring the absorbance of the extract at 451 nm using a UV-visible spectrophotometer. The formula for calculating the extraction yield is as follows:

$$X (\text{mg}/100\text{g}) = \frac{A \times V \times F \times 10}{m \times \epsilon \times b} \times 100 \quad (1)$$

where: X represents the amount of carotenoids extracted from the sample (mg/100 g); A denotes the absorbance value of the solution; V indicates the extraction volume; F signifies the dilution factor; m denotes the sample mass (g); ϵ represents the average extinction coefficient of the standard carotenoid ($2500 \text{ L}\cdot\text{mol}^{-1}\cdot\text{cm}^{-1}$); b indicates the cuvette thickness (1 cm).

2.6. Response surface optimization

The single-factor experiments conducted using the DES screening method are detailed in the supplementary materials (Fig. S2). To further enhance extraction efficiency, we integrated the results from these single-factor experiments and employed a Box-Behnken (BBD) central composite design. Response surfaces were designed using Design-Expert 13.0 software. The Response Surface Methodology (RSM) was employed with the objectives of optimizing the experimental design, evaluating the interactions among the factor variables, and minimizing the number of trials required to optimize three variables: extraction temperature, liquid-to-solid ratio, and extraction time. A three-factor, three-level experiment was conducted, with each factor level replicated three times. The factors and levels for the response surface experiment are shown in Table 1.

2.7. Antioxidant activity assay of carotenoid extracts

The antioxidant capacity of the carotenoid extract was assessed employing DPPH and ABTS free radical scavenging assays. The methods were adapted from Li et al. (2022) with modifications. In the DPPH assay, 1 mL of the sample extract was prepared with 3 mL of a 0.1 mM DPPH solution. After allowing the mixture to react in darkness for 30 min, the absorbance was recorded at 517 nm. For the ABTS assay, the radical cation was generated by reacting 7.4 mM ABTS solution to an equal volume of 2.45 mM potash peroxydisulfate solution and preserving the mixture in the dark at room temperature for 16 h. Before analysis, the stock solution was divided into aliquots and diluted with anhydrous ethanol to an absorbance of 0.70 ± 0.02 . Then, it was taken 0.5 mL of the extract and mixed with 4 mL of diluted ABTS solution, incubated for 30 min under dark conditions, and the absorbance was measured at 734 nm. A solution of ascorbic acid (Vc, 0.17 mM), prepared by dissolving

Table 1
The factors and levels table of response surface experiment design.

Factor	Level		
	-1	0	1
A: Extraction temperature ($^{\circ}\text{C}$)	30	40	50
B: liquids-to-solid ratio (mL/g)	5:1	10:1	15:1
C: Extraction time (min)	10	15	20

3.0 mg in 100 mL of water, was utilized as a positive control in both assays. The radical scavenging rate is calculated using the following formula.

$$\text{Free Radical Scavenging Rate (\%)} = \left[1 - \left(\frac{A_1 - A_2}{A_0} \right) \right] \times 100 \quad (2)$$

where: A_0 , A_1 , and A_2 represent the DPPH or ABTS absorbance values for the control group, sample group, and blank group, respectively.

2.8. Stability testing of carotenoid compounds

Determination of the effect of different solvents (deep eutectic solvent, methanol, and acetone) on the stability of extracted carotenoid compounds under varying temperatures and light exposure conditions. Extract solutions in different solvents were heated in water baths at 40°C and 80°C , with samples taken every 20 min. After rapid cooling, absorbance at 451 nm was measured. Extract solutions in different solvents were placed under alternating light (day-night cycles) and dark conditions for varying durations, followed by absorbance measurement at 451 nm.

$$\text{Carotenoid Retention Rate (\%)} = \frac{A_0}{A_1} \times 100\% \quad (3)$$

where: A_0 represents the carotenoid content after treatment; A_1 represents the carotenoid content before treatment.

2.9. The pH-switchable properties of DES

Distilled water was added to DES at varying ratios to form a two-phase system. Following the method of Chen et al. (2023), the pH of the DES was adjusted by adding polyetheramine D-230 to obtain a homogeneous phase, which was then refrigerated at 4°C for 24 h. Subsequently, hydrochloric acid solution was added to the homogeneous mixture to adjust the pH. Ultimately, the critical pH values for the switchable solvent between DES and water were determined at different volume ratios.

2.10. Fourier transform infrared spectroscopy (FTIR) analysis

FTIR analysis was performed using a Nicolet IS50 instrument (Thermo Nicolet Company, USA) on decanoic acid (C10), lauric acid (C12), DES (C10:C12 = 2:1), polyetheramine D-230, DES-D-230, and DES (switched back to hydrophobic). After 32 scans with air as the background, spectrum was collected in the wavelength region of 4000 cm^{-1} to 500 cm^{-1} at a resolution of 4 cm^{-1} .

2.11. DES-switched carotenoid recycling

A measured volume of DES extract was placed in a test tube and mixed with deionized water at a 1:3 (v/v) ratio to establish a biphasic system. Polyetheramine D-230 was then added, and the mixture was shaken vigorously to achieve a single hydrophilic phase through in situ solvent switching. The resulting solution was immersed in an ice-water bath for 2 min, followed by storage at room temperature in the dark for 48 h. Centrifuge the sample (5000 rpm, 10 min) to separate the precipitated carotenoids, which are then analyzed by spectrophotometry.

$$\text{Recovery rate (\%)} = \frac{S}{L} \times 100\% \quad (4)$$

where: S represents the carotenoid content in the precipitate, and L represents the carotenoid content in the hydrophobic DES.

2.12. Reuse of DES

The reusability of the pH-switchable DES was assessed over five consecutive cycles. After each extraction, the DES was regenerated by pH switching and carotenoid precipitation, and then reused in the next cycle. The solvent recovery rate and carotenoid extraction yield were measured after each cycle to evaluate the stability and reusability of the DES system. ^1H NMR spectra of the fresh and reused DES were recorded on a Bruker AV III 400 MHz or AV NEO 400 MHz nuclear magnetic resonance spectrometer (Bruker, Switzerland), using deuterated chloroform (CDCl_3) as the solvent. Chemical shifts were reported in ppm with the residual solvent peak of CDCl_3 as the internal reference. Using the same method as in Section 2.10, the Fourier transform infrared spectra of fresh DES and reused DES were measured. The results are shown in Fig. S3 in the supplementary materials.

2.13. Preparation of nanoemulsions

A nanoemulsion with a total mass of 100 g was prepared by combining 10 g of a DES extract containing oil-phase carotenoids, 5 g of Tween 80 surfactant, and 85 g of an aqueous phase (10 mM sodium phosphate buffer, pH 7.0). A coarse emulsion was formed by homogenizing the mixture for 2 min at 10,000 rpm with the high-speed homogenizer (KM-S10, Shanghai, China). The resulting coarse emulsion was then subjected to sonication for 15 min in an ultrasonic cleaner (S6103-01, Shanghai, China; AC 220–240 V, 50 Hz; ultrasonic frequency: 40 kHz; ultrasonic power: 100 W) to achieve redispersion and form the final nanoemulsion. It should be noted that the present study employed an ultrasonic bath rather than a probe-type sonicator. These ultrasonic systems differ in energy transmission mode, cavitation intensity, and processing behavior. In probe sonication, ultrasonic energy is delivered directly into the sample medium, generally resulting in stronger localized cavitation and higher energy density. In contrast, ultrasonic baths provide a relatively milder and more homogeneous ultrasonic field, although the effective energy transferred to the sample is usually lower. Therefore, the processing performance and optimized conditions obtained using an ultrasonic bath may not be directly comparable to those achieved with probe sonication. Similar differences between ultrasonic systems and their influence on the extraction and processing of bioactive compounds have been reported in previous studies, including flavonoid extraction from *Penthorum chinense* and crocin extraction from *Gardenia jasminoides* using deep eutectic solvent-based ultrasonic systems (Huang et al., 2022; Huang, Ying, et al., 2025).

2.14. Effect of surfactant concentration on nanoemulsion formation

Following the method of Zeng et al. (2021), systems with surfactant concentrations of 1%, 3%, 5%, 7%, and 9% were prepared. A high-speed homogenizer was used to homogenize the solutions at 10,000 rpm for 2 min to produce crude emulsions. The solutions were then placed in an ultrasonic cleaner and sonicated for 15 min to achieve redispersion.

2.15. Characterization of nanoemulsions

2.15.1. Identification of nanoemulsions

An appropriate amount of the emulsion sample was collected for analysis. Several drops of an oil-soluble dye (Sudan Red III) and a water-soluble dye (methylene blue) were mixed to the sample. The diffusion rates of the droplets within the emulsion and the resulting color changes were then observed. Take an appropriate amount of the emulsion sample to be tested.

2.15.2. Determination of particle size, polydispersity index (PDI), and zeta potential

The mean particle size, PDI, and the zeta potential of nanoemulsions were measured at 25 °C by dynamic light scattering (DLS) (Zetasizer

Nano ZS, Malvern Instruments, Worcestershire, UK). The liquid samples were first diluted 100-fold with deionized water. The diluted samples were then separately introduced into a vessel dedicated for particle size analysis and a Malvern Zeta electrode cell for zeta potential measurement. Particle size and zeta potential were subsequently detected employing the standard parameters of the instrument under their respective measurement modes.

2.15.3. Transmission electron microscopy (TEM) analysis

A 20 μL aliquot of the emulsion sample was dispensed onto a carbon-supported copper mesh and allowed to adsorb for 5 min. The excess solution was then carefully removed. Subsequently, the sample was minus-stained in 1% phosphotungstic acid for 2 min, after which surplus staining solution was removed. The grid was air-dried at room temperature prior to imaging. The microstructure was observed by field emission transmission electron microscopy (Talos G2 200 $\times$, USA).

2.16. Study on the stability of nanoemulsions

The stability of the nanoemulsion was evaluated from the method proposed by Wu et al. (2025), with certain modifications made accordingly. The pH of the emulsion was regulated to 3.0, 4.0, 5.0, 6.0, 7.0, 8.0, and 9.0 with 1 mol/L HCl or NaOH, as required. After standing at 25 °C for 24 h, the average particle size, PDI value, and zeta potential of the nanoemulsion were measured.

Equal volumes of sodium chloride solutions at different concentrations (100 mmol/L, 200 mmol/L, 300 mmol/L, 400 mmol/L, and 500 mmol/L) were added to equal volumes of the nanoemulsion. After standing at 25 °C for 24 h, the average particle size, PDI value, and zeta potential of the nanoemulsion were measured.

Nanoemulsions stored at 4 °C were heated in water baths at 45 °C, 65 °C, and 85 °C for 20 min. Samples were rapidly cooled to 25 °C using running water. After standing at 25 °C for an additional 2 h, the average particle size, PDI value, and zeta potential of the nanoemulsion were measured.

The light stability test followed the method described by Ordoñez-Lozada et al. (2025). The nanoemulsions were exposed to ultraviolet (UV) light using a 16 W UV lamp (Philips, G16 T5 tube) for 0, 30, 60, 120, and 240 min. The average particle size, PDI, and zeta potential of the nanoemulsion were measured at 25 °C.

The nanoemulsions were transferred to a 5 mL sealed vial and stored at room temperature for 15 days. Measurements were taken every three days; the average particle size, PDI, and zeta potential of the nanoemulsion were measured at 25 °C.

2.17. Determination of rheological properties of nanoemulsions

The rheological properties of the emulsion were established with a rotational rheometer (MARS 60, Germany). A 40 mm stainless steel parallel plate cell with a 1 mm gap was selected, equipped with a temperature control system to ensuring all of the tests were conducted at the same temperature, which was held constant at 25 °C. The apparent viscosity of the emulsion samples was measured over a span of the range from 0.1 s^{-1} to 100 s^{-1} . The Ostwald-de Waele model was applied for fitting to quantify the shear behavior of the emulsions, yielding specific rheological parameters K and n. Oscillatory rheological measurements were carried out by frequency sweep over the range of 0.1–10 Hz at 25 °C, and the storage modulus (G') and loss modulus (G'') were recorded to evaluate the viscoelastic properties of the nanoemulsion.

$$\eta \text{ (Pa}\cdot\text{s)} = K\gamma^{n-1} \quad (5)$$

where: η (Pa·s) is the apparent viscosity, K (Pa·sⁿ) is the consistency index, γ (s⁻¹) is the shear rate, and n (dimensionless) is the flow property index.

2.18. Simulated gastrointestinal digestion

Simulated gastrointestinal digestion was performed according to the method described by Wang et al. (2026) with minor modifications. Oral phase: Nanoemulsion (10 mL) was mixed with 10 mL of simulated salivary fluid (SSF), and the pH was adjusted to 7.0. The mixture was incubated at 37 °C under continuous agitation, and 2 mL aliquots were collected at 2, 4, 6, 8, and 10 min. Gastric phase: The entire oral digesta (20 mL) was mixed with 20 mL of simulated gastric fluid (SGF), and the

pH was adjusted to 2.5. The mixture was incubated at 37 °C under continuous agitation, and 2 mL samples were collected at 30, 60, 90, and 120 min. After digestion, the pH was neutralized to 7.0 to terminate gastric digestion. Intestinal phase: The gastric digesta was mixed with 20 mL of simulated intestinal fluid (SIF), and the pH was adjusted to 7.0. The mixture was incubated at 37 °C under continuous agitation, and 2 mL samples were collected at 150, 180, 210, 240, 270, 300, 330, and 360 min. After each sampling, an equal volume (2 mL) of fresh SSF, SGF, or SIF was added to maintain a constant volume. The absorbance of each

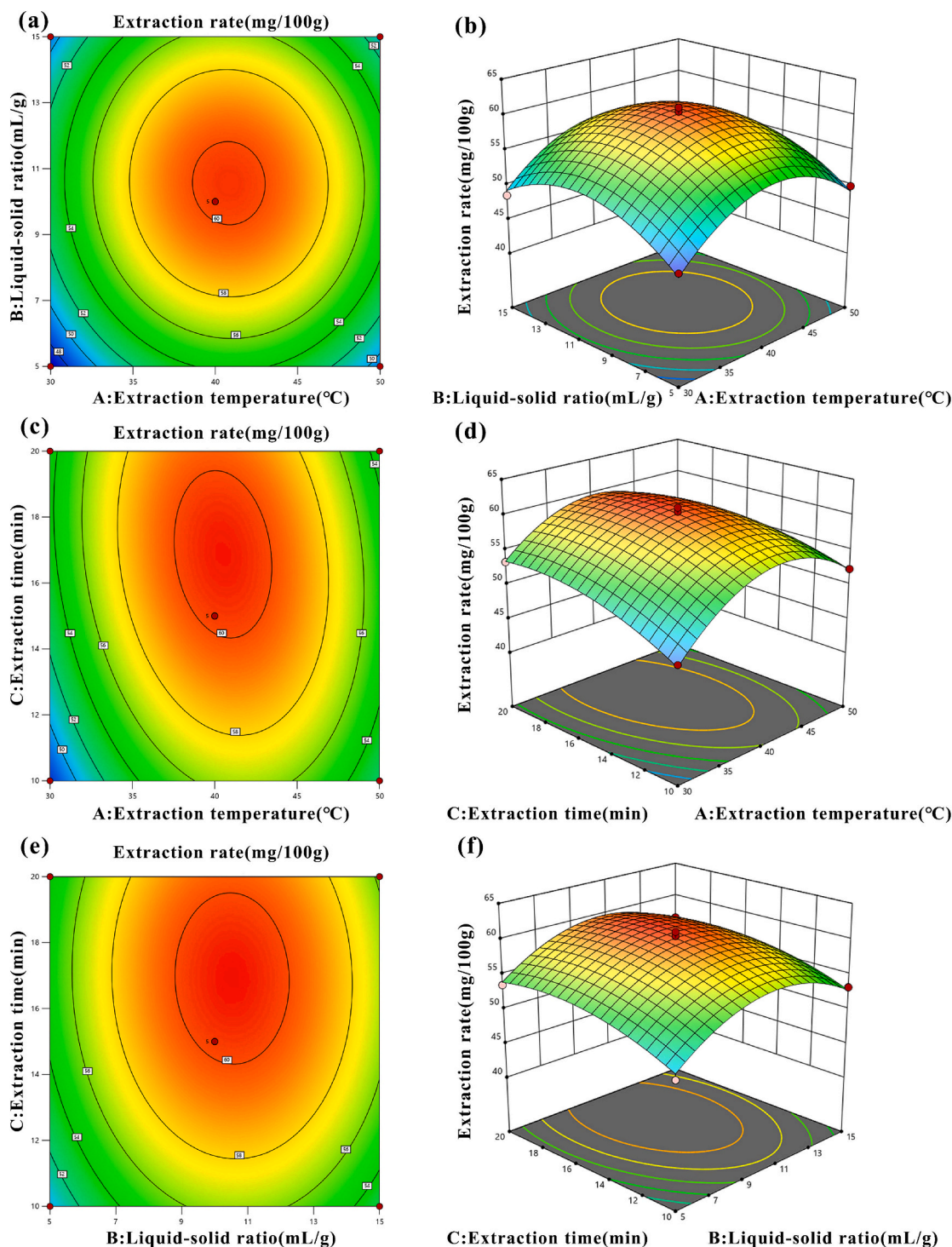


Fig. 1. Contour diagram (a, c, e) and 3D response surface model diagram (b, d, f) of the interaction between extraction time, extraction temperature and liquid-solid ratio.

sample was measured at 451 nm using a UV–visible spectrophotometer (UV2500PC, Japan).

2.19. Statistical analysis

The results are presented as mean $\pm$ standard deviation ($n = 3$). Graphs were produced with Origin (2018) software. One-way analysis of variance (ANOVA) and Duncan's multiple range test were performed using SPSS 24.0 (IBM, USA). Statistical significance was set at $p < 0.05$.

3. Results and discussion

3.1. Response surface extraction and optimization

Based on the principles of BBD experimental design, an RSM experiment was conducted with extraction temperature, liquid-to-solid ratio, and extraction time as independent variables, and extraction rate as the response value. Table S2 presents the BBD experimental design scheme and results for three factors at three levels, where Y represents the carotenoid extraction rate. Fig. 1 shows ontour diagram and 3D response surface model diagram of the interaction between extraction time,

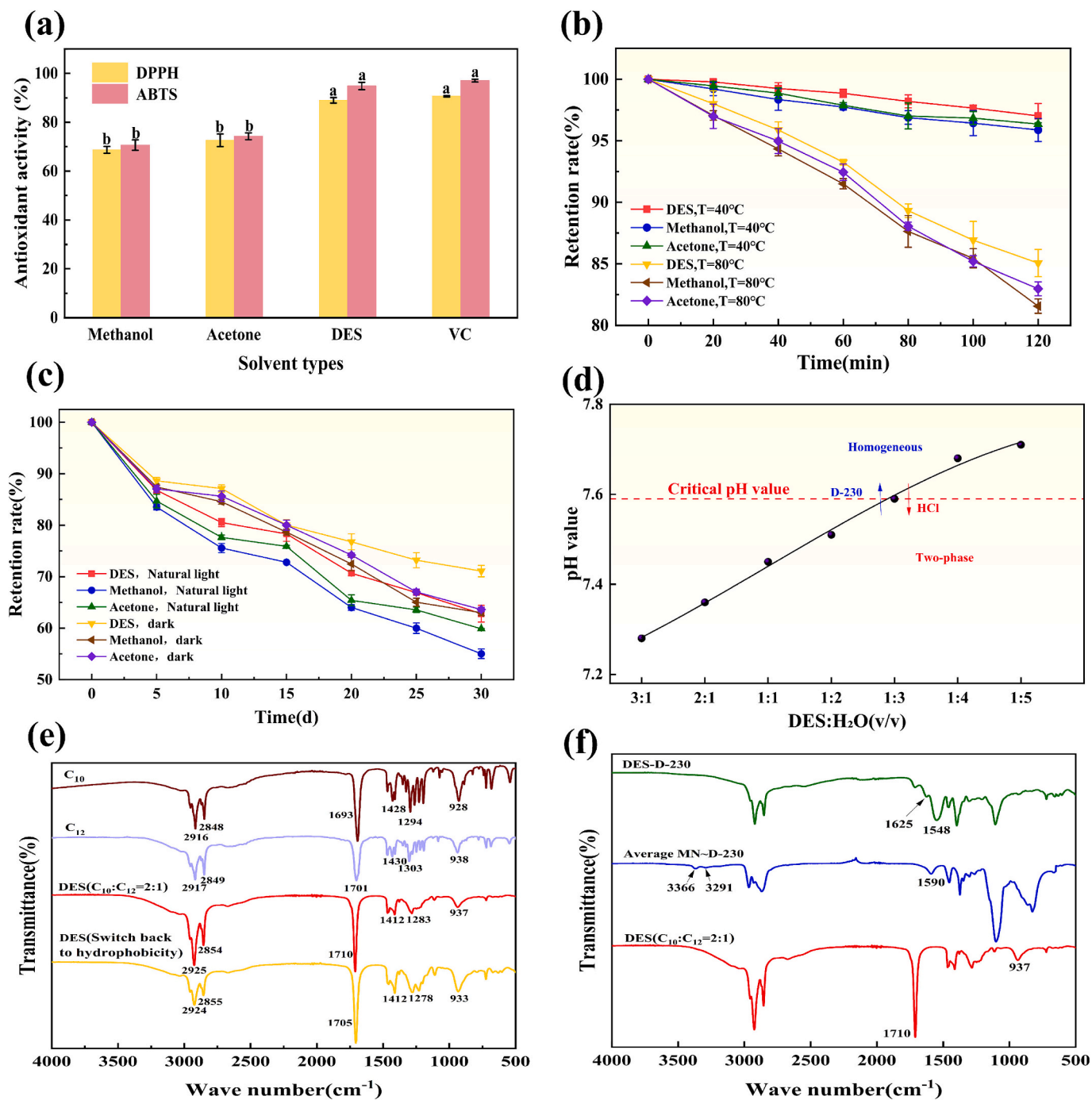


Fig. 2. The antioxidant activity of the different solvents was determined by DPPH and ABTS methods (a). Stability of carotenoid in DES, Methanol and Acetone at 40 °C and 80 °C (b). Stability of carotenoid in DES, Methanol and Acetone in dark and light (c). Critical pH value of DES at different volume ratios of DES and water (d). Fourier transform infrared spectra of C₁₀, C₁₂, DES (C₁₀:C₁₂ = 2:1), D-230, DES-D-230 and DES (switching back to hydrophobicity) (e, f).

extraction temperature and liquid-solid ratio.

Using carotenoid extraction rate as the response variable, Design-Expert 13.0 software was applied to perform fitting tests on the data in the above table. A quadratic regression response analysis was conducted to establish the model equation. The regression equation describes the actual relationship between each factor and Y, enabling the determination of optimal conditions during the extraction process for theoretical prediction of test results. Based on the above analysis, the quadratic equation is as follows:

$Y = 60.20 + 1.07A + 1.11B + 1.68C - 0.2375AB - 1.16AC - 0.2100BC - 6.36A^2 - 4.86B^2 - 2.16C^2$, where Y represents the extraction rate of carotenoids, and A, B, and C denote the three independent variables from the table above.

Based on the significance of experimental results, regression models, and ANOVA findings (Table S3 and Table S4), F-value analysis indicates the significance of individual factors affecting extraction. The magnitude of influence on carotenoid extraction rate from the three factors is ranked as follows: C ($F = 28.16$) > B ($F = 12.25$) > A ($F = 11.48$). This demonstrates that the extraction rate is primarily influenced by extraction time, followed by liquid-to-solid ratio and extraction temperature. This indicates that the order of influence on extraction yield is: extraction time > liquid-to-solid ratio > extraction temperature.

Using Design Expert 13 software, numerical optimization methods were employed to refine the DES-based extraction system, ultimately determining an optimal set of conditions: extraction temperature of 40.49 °C, liquid-to-solid ratio of 10.52:1 mL/g, and extraction time of 16.85 min. At these conditions, the withdrawal rate reached 60.60 mg/100 g. Experiments conducted under the predicted optimal conditions allowed comparison of expected versus experimental values to evaluate model accuracy. Response surface results were adjusted based on actual experimental conditions. Experimental settings included extraction temperature of 40.50 °C, liquid-to-solid ratio of 10.5:1 mL/g, and extraction time of 16.80 min. Three validation experiments were conducted, yielding an extraction rate of 60.98 ± 0.87 mg/100 g. This value did not show remarkable variation compared to the predicted value ($p > 0.05$), indicating that the model is feasible for predicting extraction conditions for carotenoids in orange peel.

3.2. Antioxidant activity of carotenoid extracts

The biological activity of carotenoids is primarily determined by their solvent environment, making it necessary to investigate their antioxidant activity in different solvents (Li et al., 2022). The antioxidant potential of the DES extract was experimentally determined and compared with methanol and acetone extracts, with results shown in Fig. 2a. DPPH and ABTS radical scavenging assays revealed that the DES extract exhibited significantly higher antioxidant capacity than methanol and acetone extracts, with no remarkable difference compared to Vc ($p > 0.05$). Furthermore, the DES components may assist to enhancing the overall antioxidant activity of the fractions, with synergistic effects arising between the solvent and bioactive compounds, leading to elevated antioxidant potency.

3.3. Stability testing of carotenoid compound extracts

Fig. 2b shows the degradation curves of carotenoids in different solvents at 40 °C and 80 °C. At 40 °C, the retention rate of carotenoid compounds in DES decreased from 99.78% to 97.02%, and the degradation curve of carotenoid compounds extracted with DES solvent was similar to those of methanol and acetone solvents. At 80 °C, the retention rate in DES decreased sharply from 98% to 85.06%, indicating that higher temperatures accelerate the degradation of carotenoid compounds. At the extraction temperature of 80 °C, the retention rate of carotenoid compounds in DES solution was higher than that in methanol and acetone solvents, indicating greater stability of carotenoid compounds in DES solution compared to methanol and acetone solutions.

Previous studies have indicated that DESs potentiate the stabilization of biological active complexes (Stupar et al., 2021). Owing to hydrogen bonding effects between the solvent and subject compound, this interaction reduces the mobility of solute molecules and restricts their contact with oxygen at the interface between DESs and air, thereby minimizing the most common form of oxidative degradation (Cvjetko Bubalo et al., 2018). Accordingly, the stability of carotenoids in selected DESs was monitored for 30 days under light (day-night cycles) and dark conditions, with comparisons made to methanol and acetone. Results are shown in Fig. 2c. In all circumstances, carotenoids degraded faster under light than in darkness. Under light storage conditions, the retention rate in DES decreased from 86.79% to 62.79% after 30 days. In dark storage conditions, carotenoids in DES were more stable, with the retention rate decreasing from 88.64% to 71.09% after 30 days. These findings be attributed to the isomerization and oxidation processes of carotenoids (Sachindra & Mahendrakar, 2010). In both situations, the carotenoids in DES were more stable than those in methanol and acetone extracts. Similar results were also obtained in the study by Viñas-Ospino et al. (2023).

3.4. The pH-switchable properties of DES

By adjusting the pH of the solution system to alter the properties of the DES, a reversible phase transition is induced, thereby achieving the conversion between hydrophilicity and hydrophobicity (Hu et al., 2023). The critical pH values for phase inversion between DES and water were determined at different volume ratios by adjusting the pH of the DES solution using polyetheramine D-230 and HCl. As shown in Fig. 2d, the critical pH increases with the proportion of water in the solution. This can be explained by the fact that the addition of neutral water (pH 7.0) raises the overall pH of the solution (Chen et al., 2023). When the volume ratio of DES to water is 1:3, the solution transitions from a two-phase system to a homogeneous phase at pH values above 7.59 after adding polyetheramine D-230. This transition is reversible upon addition of HCl solution. This phase behavior provides qualitative evidence of the critical pH range associated with the pH-responsive switching of the DES/water system.

3.5. FTIR analysis

FT-IR characterization of decanoic acid (C10), lauric acid (C12), DES (C10:C12 = 2:1), polyetheramine D-230, DES-D-230, and DES (reverted to hydrophobic) was performed to elucidate the molecular interactions and structural changes occurring in the formed DES (Md Yusoff et al., 2023).

The 1693 cm^{-1} band in the C10 spectrum corresponds to the stretching vibration of the C=O group in the carboxyl COOH group. 1294 cm^{-1} corresponds to the stretching vibration of the C-OH bond in the carboxyl group, while the range 2520 cm^{-1} – 3170 cm^{-1} belongs to the stretching vibrations of O—H bonds. This is because stronger hydrogen bonding interactions formed by O—H groups within or between molecules cause the O—H stretching vibrations to shift toward lower frequencies, resulting in broader absorption bands. In the C12 spectrum, 1701 cm^{-1} corresponds to the C=O stretching vibration of the carboxyl group, 1303 cm^{-1} to the C-OH stretching vibration on the carboxyl group, and 2500 cm^{-1} – 3200 cm^{-1} to the O—H stretching vibration. The shift in characteristic peaks in DES indicates the formation of hydrogen bond interactions. In the DES (C10:C12 = 2:1) spectrum, the C=O group in -COOH exhibits stretching vibration at 1710 cm^{-1} , while 1283 cm^{-1} corresponds to the stretching vibration of the C=O group on the carboxyl group. The O—H stretching vibration is assigned to the range of 2460 cm^{-1} to 3340 cm^{-1} . Compared to C10 and C12, the O—H stretching vibration band on the DES carboxyl group broadens to 2460 cm^{-1} – 3340 cm^{-1} , confirming successful hydrogen bond formation between components. This broadening arises from reduced force constants due to proton transfer via hydrogen bonds between components.

(Santana et al., 2019). Additionally, the shift of the C=O group from 1693 cm^{-1} and 1701 cm^{-1} to 1710 cm^{-1} represents another phenomenon indicative of hydrogen bond formation. No new peaks appeared in the prepared DES, suggesting no additional chemical reactions occurred between the components. Therefore, the shifts in the wavenumbers of the O—H and C—O groups indicate that a hydrogen bond formed between C10 and C12 near the -COOH group, confirming the successful formation of the DES.

As shown in Fig. 2f, after DES was neutralized by polyetheramine D-230, the C=O peak at 1710 cm^{-1} and the C-O-H peak at 937 cm^{-1} disappeared, while a stretching vibration peak for -COO- appeared at 1548 cm^{-1} , indicating the conversion of -COOH to -COO-D-230. As shown in the spectrum of polyetheramine D-230 in Fig. 2f, the anti-symmetric stretching vibration, symmetric stretching vibration, and bending vibration peaks of the -NH₂ group are located at 3366 cm^{-1} , 3291 cm^{-1} , and 1590 cm^{-1} , respectively. Upon neutralization of polyetheramine D-230, the peaks at 3366 cm^{-1} and 3291 cm^{-1} disappear, while the bending vibration peak at 1590 cm^{-1} shifts to 1625 cm^{-1} , indicating protonation of -NH₂ (Chen et al., 2023). Furthermore, the characteristic peaks of DES showed no significant changes before and after the pH switching, indicating that no structural changes occurred during the switching process.

3.6. Recovering carotenoids via DES switching

Current research on carotenoid separation and purification primarily focuses on the polarity tunability of fatty acid-based DES to separate carotenoids from extraction solvents. Stupar et al. (2021) synthesized a series of fatty acid-based hydrophobic DES and combined them with an ultrasonic-assisted method to extract β -carotene from pumpkin, achieving a recovery rate of 93.60%. Zhang et al. (2023) employed a combination of switchable deep eutectic solvents and phenolic antioxidants to extract β -carotene from millet. Adjusting the pH with HCl converts hydrophobic DES into hydrophilic DES, enabling the recovery of hydrophobic β -carotene solids. This process achieves direct solid-state recovery of β -carotene at rates exceeding 91.00%. The process of extracting carotenoids from orange peel using switchable DESs is shown in Fig. 3.

Based on the experimental results regarding the pH-switchable properties of 3.4 DES and the extraction capacity of fatty acid-based eutectic solvents for carotenoids in carotenoid-rich extracts, a method for the sustainable separation of carotenoids from hydrophobic fatty acid-based DES extracts was proposed. After carotenoids were extracted by the hydrophobic DES, the pH was adjusted using polyetheramine D-230, which transformed the hydrophobic DES into a hydrophilic form and promoted carotenoid precipitation, enabling their separation by centrifugation. Subsequent treatment with HCl converted the hydrophilic DES back to its hydrophobic form, allowing the solvent to be reused for carotenoid extraction from orange peel powder. The results showed a 90.30% solid-state recovery rate for carotenoids using this method.

3.7. Effect of surfactant concentration on nanoemulsion formation

As shown in Fig. 4a, the average particle size of HDES-based nanoemulsions is significantly influenced by the consistency of Tween 80. In the range of 1% to 7% concentration, the trend shows that the mean particle size of the emulsion decreases as the surfactant concentration in the system increases. At a concentration of 7%, the emulsion exhibits the smallest particle size, consistent with the findings reported by Zeng et al. (2021). This is because increasing the surfactant concentration reduces the surface tension at the oil-water interface, thereby promoting droplet formation. Additionally, higher surfactant concentrations cause surfactant molecules to cover the droplet surface more rapidly, thus better preventing droplet coalescence (Chang et al., 2013). As the Tween 80 concentration was further increased to 9%, the particle size exhibited an increase. This may be attributed to excessive surfactant concentration leading to interfacial saturation. The surplus surfactant formed micelles in the aqueous phase rather than coating the droplet surfaces, resulting in enlarged particle size and droplet aggregation (Nejadmansouri et al., 2016). The PDI value of the nanoemulsion decreases with increasing Tween 80 concentration, with a particularly significant reduction observed when the concentration increases from 5% to 7%. Furthermore, Fig. 4b shows that at Tween 80 concentrations of 1%, 3%, and 5%, the emulsion particle size distribution exhibited a bimodal profile. At concentrations of 7% and 9%, the distribution presented only a narrow

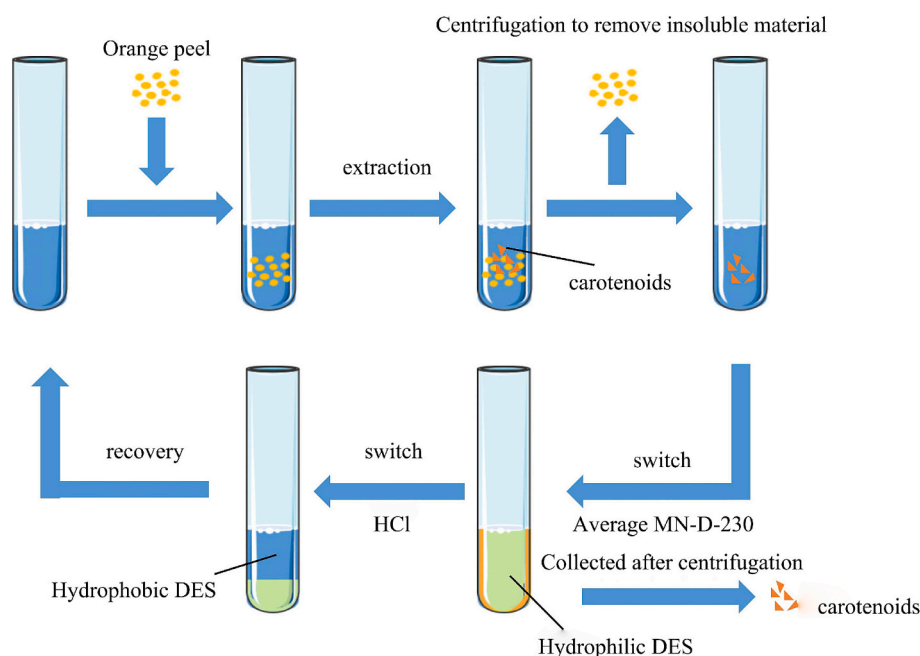


Fig. 3. The process of extracting carotenoids from orange peel using switchable DESs. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

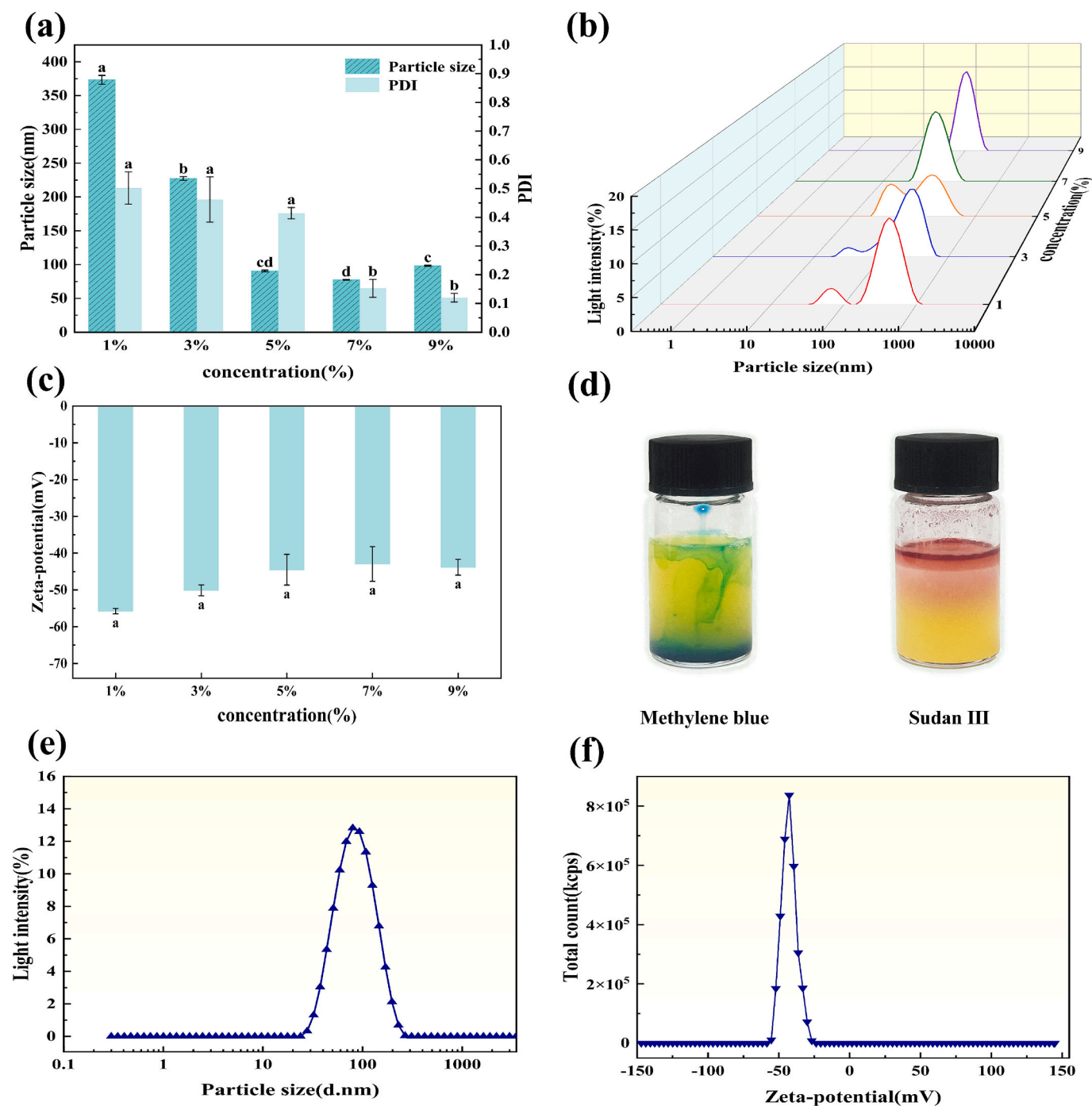


Fig. 4. Effect of Tween 80 concentration on particle size and PDI (a). Particle size distribution of nanoemulsions (b). Effect of Tween 80 concentration on zeta-potential of nanoemulsions (c). Identification of nanoemulsions type via staining method (d). Particle size and zeta potential maps of nanoemulsions (e, f).

peak, indicating a uniform particle size distribution. Research indicates that greater variation in droplet size leads to larger changes in surface energy, resulting in increased instability over time (Sadeghian et al., 2023).

Fig. 4c shows the effect of Tween 80 concentration on the zeta potential of the emulsion. The results indicate that the zeta potential of the emulsion generally increases as the concentration of Tween 80 increases. The HDES-based nanoemulsion stabilized by nonionic surfactants carries a negative charge due to the adsorption of HDES components—such as free fatty acids—on the droplet surface. At low Tween 80 concentrations, the larger size of emulsion particles provides favorable conditions for the adsorption of more negative ions onto their

surface, which consequently results in an overall increase in the zeta potential of the emulsion. With increasing concentrations of Tween 80, more Tween 80 aggregates at the droplet surface, thereby reducing the zeta potential of the emulsion. This finding is consistent with the results reported by Zeng et al. (2021). Although electrostatic repulsion may not be the primary stabilization factor for Tween 80-based nanoemulsions, surfactant strength needs to be fully considered when pursuing more consistent and cost-effective emulsions. Therefore, we selected a nanoemulsion with a 7% Tween 80 concentration for further evaluation.

3.8. Characterization of nanoemulsions

3.8.1. Identification of nanoemulsions

The staining method utilizes the principle that the water-soluble dye methylene blue (blue) diffuses rapidly in the aqueous phase, while the oil-soluble Sudan III (red) diffuses rapidly in the oil phase. By comparing the diffusion rates of these two dyes in the nanoemulsion, the emulsion type can be determined. Fig. 4d clearly demonstrates that methylene blue diffuses faster than Sudan III within the emulsion, confirming that the prepared nanoemulsion is of the O/W type. O/W emulsions hold greater practical value due to their ability to be infinitely diluted in water without structural alteration. The increased oil/water interface surface area facilitates rapid digestion within the gastrointestinal tract (Jafari & McClements, 2017).

3.8.2. Determination of particle size, PDI, and zeta potential

The particle size distribution results of the nanoemulsion prepared in this experiment are shown in Fig. 4e, with an average size of 76.78 ± 1.58 nm and a PDI of 0.15 ± 0.01 . Particle size affects the stability, optical properties, and nutrient release characteristics of nanoemulsions; generally, smaller particle sizes yield better performance (Sharma et al., 2021). A PDI value between 0.1 and 0.3 indicates a monodisperse system. A higher PDI value signifies a more complex system, while a lower PDI value indicates a more uniform particle size distribution (Mazzarino et al., 2018). PDI = 0.3 indicates that the droplets exhibit a narrow distribution with a particle size of 76.78 nm in the emulsion system, demonstrating high uniformity and belonging to a monodisperse system.

Zeta potential serves as an indicator for measuring the repulsive forces between emulsion droplets (Shanmugam & Ashokkumar, 2014). The greater the electrostatic repulsion between emulsion droplets, the less likely they are to aggregate, thereby forming a more stable system (Li, Zhao, et al., 2020). The Zeta potential of the emulsion prepared in this study is negative, indicating that the emulsion droplets carry a negative charge. Research indicates that as a nonionic surfactant, Tween 80 does not impart any charge. The negative charge on droplets stabilized by nonionic surfactants may result from free fatty acid impurities in the oil phase or adsorption of OH^- of the water onto the droplet surface (Ziani et al., 2011). Since the components of HDES in this study contain easily ionizable compounds such as free fatty acids, these negative charges may result from the adsorption of HDES components on the droplet surface. For charge-stable nanoemulsions, the greater the absolute value of the zeta potential, the more stable the system. Generally, a zeta potential exceeding +30 mV or below -30 mV indicates long-term stability of the nanoemulsion (Borthakur et al., 2016). As shown in Fig. 4f, the zeta potential value of the HDES-based nanoemulsion is -42.56 ± 1.59 mV, indicating good stability.

3.8.3. TEM analysis

TEM analysis was performed on the nanoemulsion to observe its size and surface morphology. As shown in Fig. 5, the nanoemulsion droplets exhibit spherical or ellipsoidal shapes with good formation properties. They are uniformly dispersed within the field of view without any aggregation. The particle size obtained from TEM is less than 100 nm, consistent with the results from dynamic light scattering. The spherical morphology indicates appropriate homogenization of the emulsion, with good mixing between the surfactant and oil phase, confirming the formation of a stable nanoemulsion.

3.9. Stability study of nanoemulsions

3.9.1. Effect of pH on the stability of nanoemulsions

Fig. 6a displays the impact of various pH values on the particle size and PDI of the nanoemulsion. With increasing pH, the particle size of the emulsion first decreases and then increases. At pH values between 3 and 7, the emulsion particle size remains below 150 nm, meeting the criteria

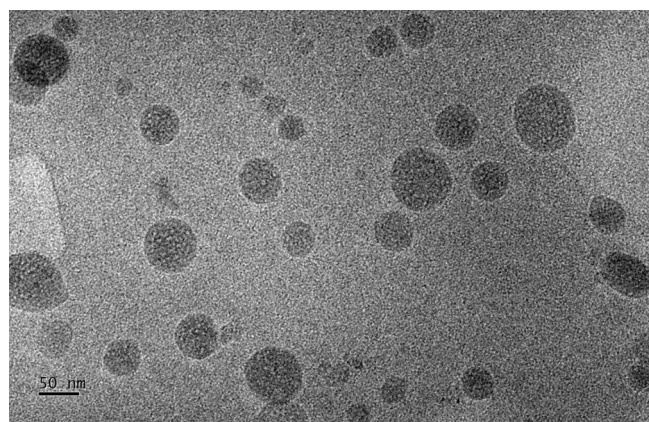


Fig. 5. TEM plots of nanoemulsions.

for nanoemulsions. The PDI value increases with rising pH. At pH 8 and 9, both particle size and PDI significantly increase, indicating structural disruption of the nanoemulsion under alkaline conditions. HDES-based emulsions composed of fatty acids are significantly affected by pH, likely due to the reactivity of fatty acids with alkaline solutions, consistent with the findings of Zeng et al. (2021).

As shown in Fig. 6b, we investigated the effect of pH on the emulsion. In the pH range of 3 to 7, the absolute number of zeta potential reaches its maximum at pH 5, indicating excellent emulsion stability at this pH value. At pH 8 and pH 9, the absolute values of the zeta potential were significantly higher than those observed in the pH 3–7 range. This may occur because free fatty acids preferentially adsorb onto droplet surfaces under high pH conditions. Based on these results, the prepared nanoemulsion is susceptible to disruption under alkaline conditions. Therefore, alkaline environments should be avoided during food processing and storage. Additionally, the prepared nanoemulsion can effectively protect carotenoids from the highly acidic environment of the stomach.

3.9.2. Effect of NaCl concentration on the stability of nanoemulsions

In food production, NaCl or other salts are commonly added for multiple purposes. Delivery systems based on nanoemulsions can be utilized to prepare functional food and beverage products containing salt ions (Zhao et al., 2022). As shown in Fig. 6c, as the NaCl concentration gradually increased from 0 to 500 mmol/L, the particle size of the nanoemulsion exhibited a continuous and significant trend of enlargement. At a NaCl concentration of 0, the particle size was approximately 70 nm; when the concentration rose to 500 mmol/L, the particle size substantially increased to about 280 nm. Significant differences in particle size were observed across varying NaCl concentrations, indicating that increasing NaCl concentration promotes aggregation of nanoemulsion droplets, leading to larger particle sizes. The trend in PDI differed from that of particle size. As NaCl concentration increased from 0 to 100 mmol/L, PDI slightly increased; but when the concentration further increased from 200 mmol/L to 500 mmol/L, the PDI decreased significantly. This change can be attributed to the shielding effect of NaCl on the electrostatic repulsion at the surface of nanoemulsion droplets under high ionic strength. This shielding reduces the electrostatic repulsion between droplets to a level insufficient to overcome van der Waals forces and hydrophobic interactions, leading to droplet aggregation. Consequently, the particle size distribution becomes more uniform, resulting in a decrease in PDI (Bai et al., 2024).

As shown in Fig. 6d, at different NaCl concentrations, the absolute value of the zeta potential of the emulsion decreases as NaCl concentration increases. Without added NaCl, the zeta potential of the prepared nanoemulsion was -38.72 mV. Increasing the NaCl concentration to 500 mmol/L further reduced the zeta potential value to -23.05 mV. Adding more NaCl drives more Na^+ ions to neutralize the negative charge on the droplet surface. Consequently, the electrophoretic

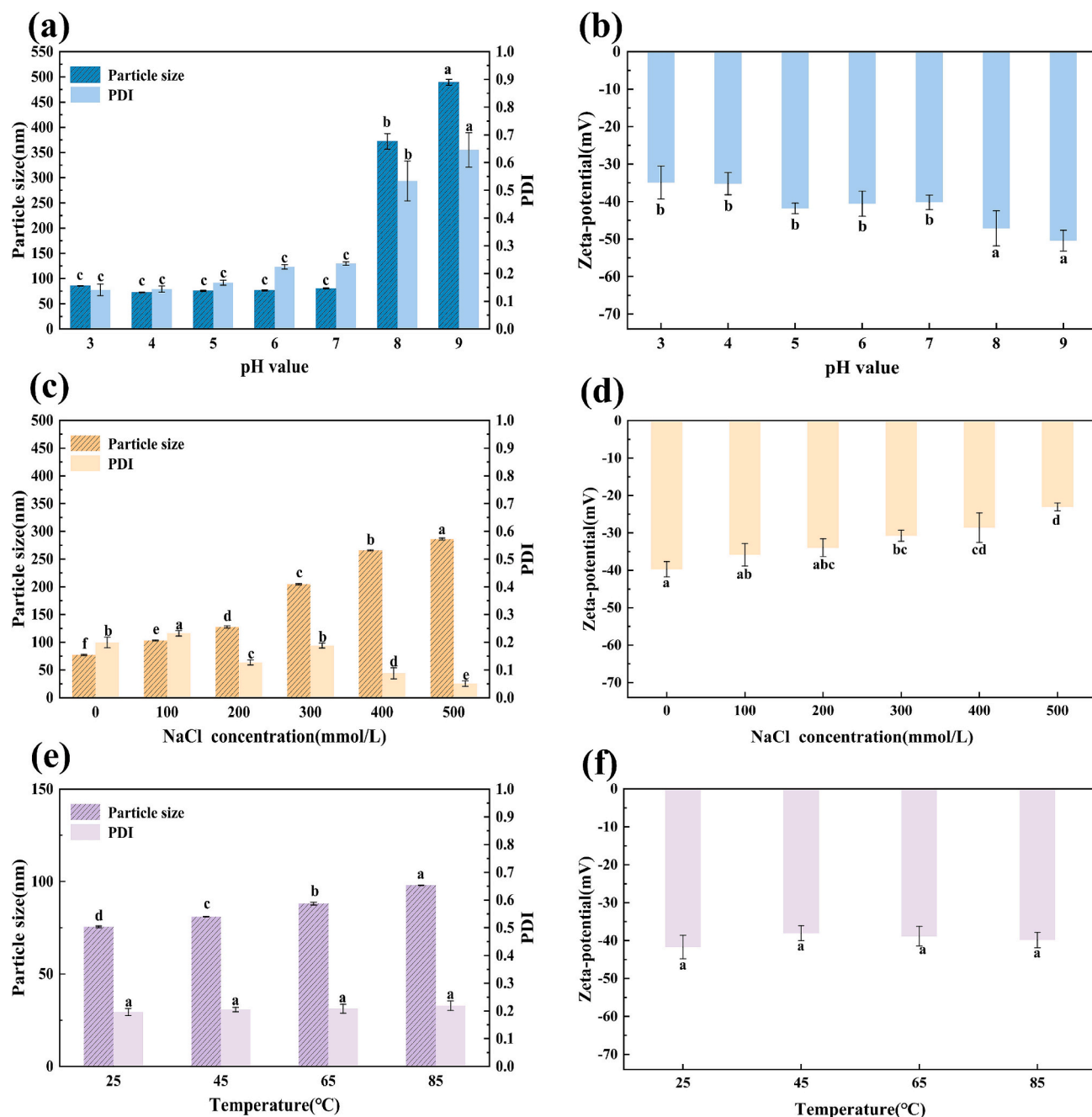


Fig. 6. Effect of pH on the particle size, PDI (a), and zeta potential (b) of nanoemulsions. Effect of NaCl concentration on particle size, PDI (c), and zeta potential (d) of nanoemulsions. Effect of temperature on particle size, PDI (e), and zeta potential (f) of nanoemulsions.

mobility and electric field strength of the nanoemulsion droplets decrease, leading to a reduction in the absolute zeta potential (since zeta potential is proportional to electrophoretic mobility) (Li et al., 2018).

3.9.3. Effect of heat treatment on the stability of nanoemulsions

As shown in Fig. 6e, after heating for 20 min at temperatures of 45 °C, 65 °C, and 85 °C, the particle size of the emulsion increased significantly with rising temperature, while the PDI value increased but not significantly ($p>0.05$). This result may be ascribed to interfacial reactions between droplets. Furthermore, the magnitude of droplet size increase was greater at higher heating temperatures, consistent with the findings of Yang et al. (2023). However, across the entire temperature range, the particle size of the emulsion remains below 100 nm, meeting the specified particle size for composite nanoemulsions.

As shown in Fig. 6f, during the heat treatment process, the zeta potential showed no significant change with increasing temperature, indicating that heating had no pronounced effect on the structure of the nanoemulsion. Although elevated ambient temperatures may reduce the stability of nanoemulsions, the nanoemulsion prepared in this experiment remained relatively stable under certain heating conditions, making it a potential candidate material for application in food emulsion systems.

3.9.4. Effect of light exposure on the stability of nanoemulsions

As shown in Fig. 7a, the particle size of the nanoemulsions remained relatively stable during the initial storage period under both UV irradiation and light-protected conditions, with only a slight increase observed over time. After 240 min, the particle size of all formulations

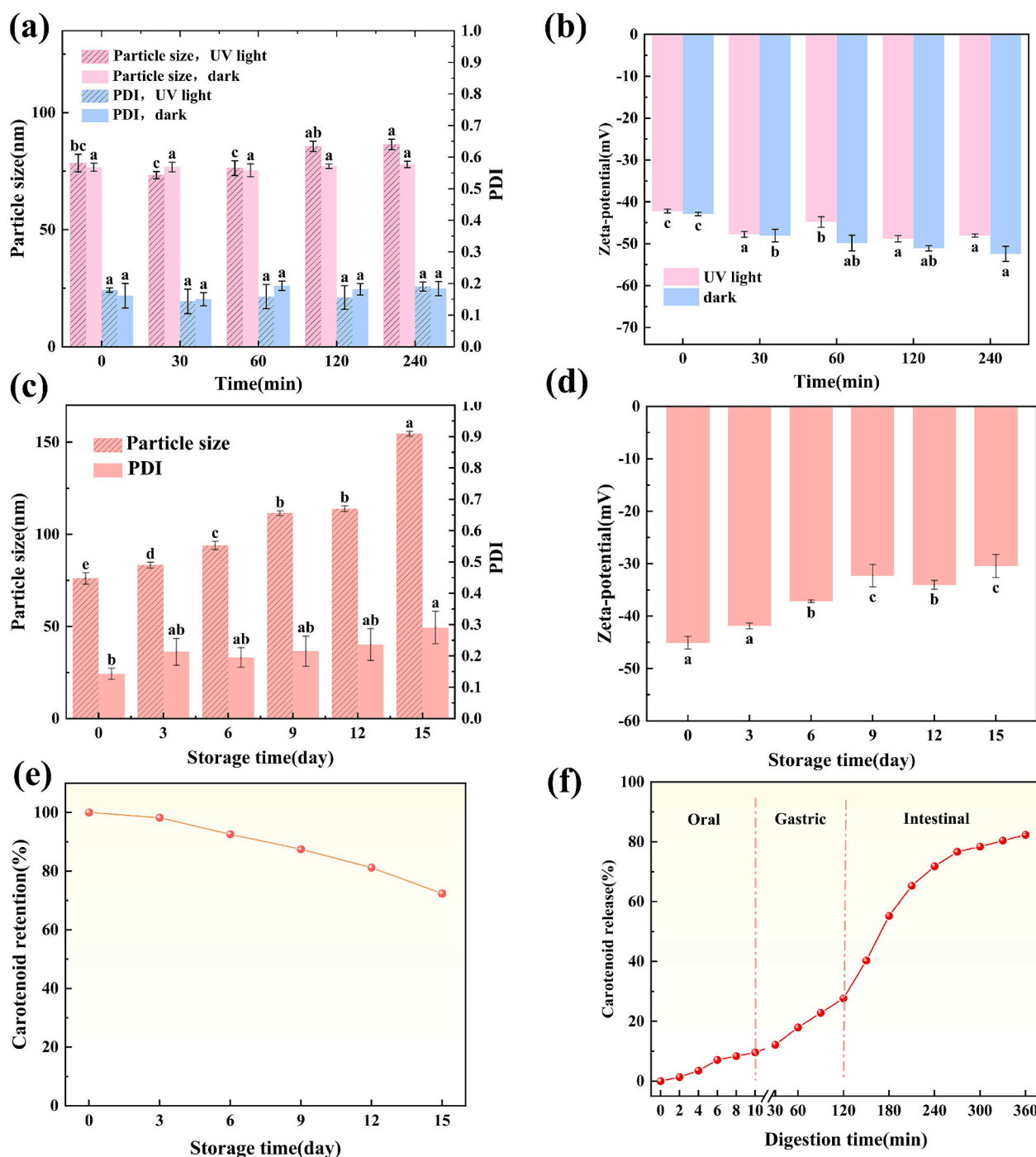


Fig. 7. Effect of light exposure on the particle size, PDI (a), and zeta potential (b) of nanoemulsions. Effect of storage time on particle size, PDI (c), and zeta potential (d) of nanoemulsions. Retention rate of carotenoids during storage time (e). Carotenoids release during simulated gastrointestinal digestion (f).

remained below 100 nm, indicating that the nanoemulsions retained their nanoscale characteristics. Although UV irradiation appeared to induce a slightly greater increase in particle size than storage in the dark, the overall change was limited, suggesting that the system was able to resist severe aggregation or droplet swelling during short-term light exposure. Furthermore, the PDI remained below 0.2 throughout the experiment, confirming that all nanoemulsions maintained a narrow particle size distribution and good monodispersity.

As shown in Fig. 7b, the zeta potential became increasingly negative over time under UV irradiation. This trend indicates that the surface charge of the droplets increased during the test, which may have enhanced electrostatic repulsion between droplets and thereby

contributed to colloidal stability (Mudalige et al., 2019). This increase may be attributed to the adsorption of charged degradation products generated during UV irradiation onto the droplet surfaces, thereby increasing the surface charge density (Ordoñez-Lozada et al., 2025).

3.9.5. Effect of storage time on the stability of nanoemulsions

As shown in Fig. 7c and d, with the extension of storage time, the particle size of the nanoemulsions increased gradually, while the absolute value of zeta potential decreased, indicating a gradual reduction in colloidal stability during storage. The increase in particle size may be attributed to droplet aggregation or Ostwald ripening during storage, whereas the decrease in zeta potential suggests weakened electrostatic

repulsion between droplets.

As shown in Fig. 7e, the carotenoid retention rate decreased continuously over the storage period, which may be related to the susceptibility of carotenoids to oxidation and degradation during storage, as well as the reduced protective effect of the nanoemulsion system when its physical stability declined (Su et al., 2025). Overall, although the nanoemulsion remained relatively stable in the short term, prolonged storage led to increased particle size, reduced surface charge, and lower carotenoid retention. Nevertheless, the system still retained a considerable proportion of carotenoids after 15 days, suggesting its potential application in food delivery systems.

3.10. Rheological behavior of nanoemulsions

In the 3.9 stability study, the relationship between nanoemulsion stability and particle size, PDI value, and zeta potential has been elucidated. Another significant characteristic of nanoemulsions is their fluidity (e.g., viscosity) (Foudazi et al., 2015), hence shear rate scans were conducted on nanoemulsions at different pH levels, NaCl concentrations, and after heat treatment. As shown in Fig. 8, starting from 0.1 s^{-1} and with increasing shear rate, the viscosity of the emulsion gradually decreases. All samples exhibit shear thinning behavior, indicating that the samples are pseudoplastic fluids. Prior to shear application, O/W nanoemulsion droplets arrange themselves in a manner that reduces the Gibbs free energy that is associated with the system. Upon shear application, droplets separate and align along the flow direction, diminishing flow resistance and consequently lowering viscosity (Ariffin

et al., 2016).

Fig. 8a illustrates the impact of pH on the apparent viscosity of nanoemulsions. The figure indicates that samples prepared at pH 5 and pH 6 exhibit good parallelism with the control group. The pH of the nanoemulsion prepared in this study was 5.19, suggesting that minor pH adjustments have a negligible impact on emulsion viscosity. However, samples at pH 8 and pH 9 exhibit significantly reduced apparent viscosity. This occurs because the acidic nature of the oil phase in the nanoemulsion causes oil droplets bound within the network to leak out when pH is adjusted to alkaline conditions. Consequently, viscosity decreases, leading to demulsification and reduced stability (Guo et al., 2022).

Relevant data were analyzed using the power-law model. As shown in Table 2, the nanoemulsion exhibited a flow behavior index n below 1 across the pH range of 3–9, indicating it is a typical pseudoplastic fluid.

Table 2

Herschel-Bulkley model parameters of nanoemulsions with different pH values.

pH	K (Pa·s ⁿ)	n	R ²
CK	25.42 ± 0.86	0.4391 ± 0.0201	0.9371
3	32.79 ± 1.11	0.5002 ± 0.0204	0.9116
4	38.23 ± 0.77	0.6282 ± 0.0126	0.9299
5	29.84 ± 0.97	0.4261 ± 0.0192	0.9461
6	27.41 ± 0.96	0.4123 ± 0.0207	0.9409
7	19.51 ± 0.69	0.4415 ± 0.0209	0.9314
8	15.73 ± 0.44	0.4587 ± 0.0167	0.9530
9	12.18 ± 0.41	0.5319 ± 0.0206	0.8946

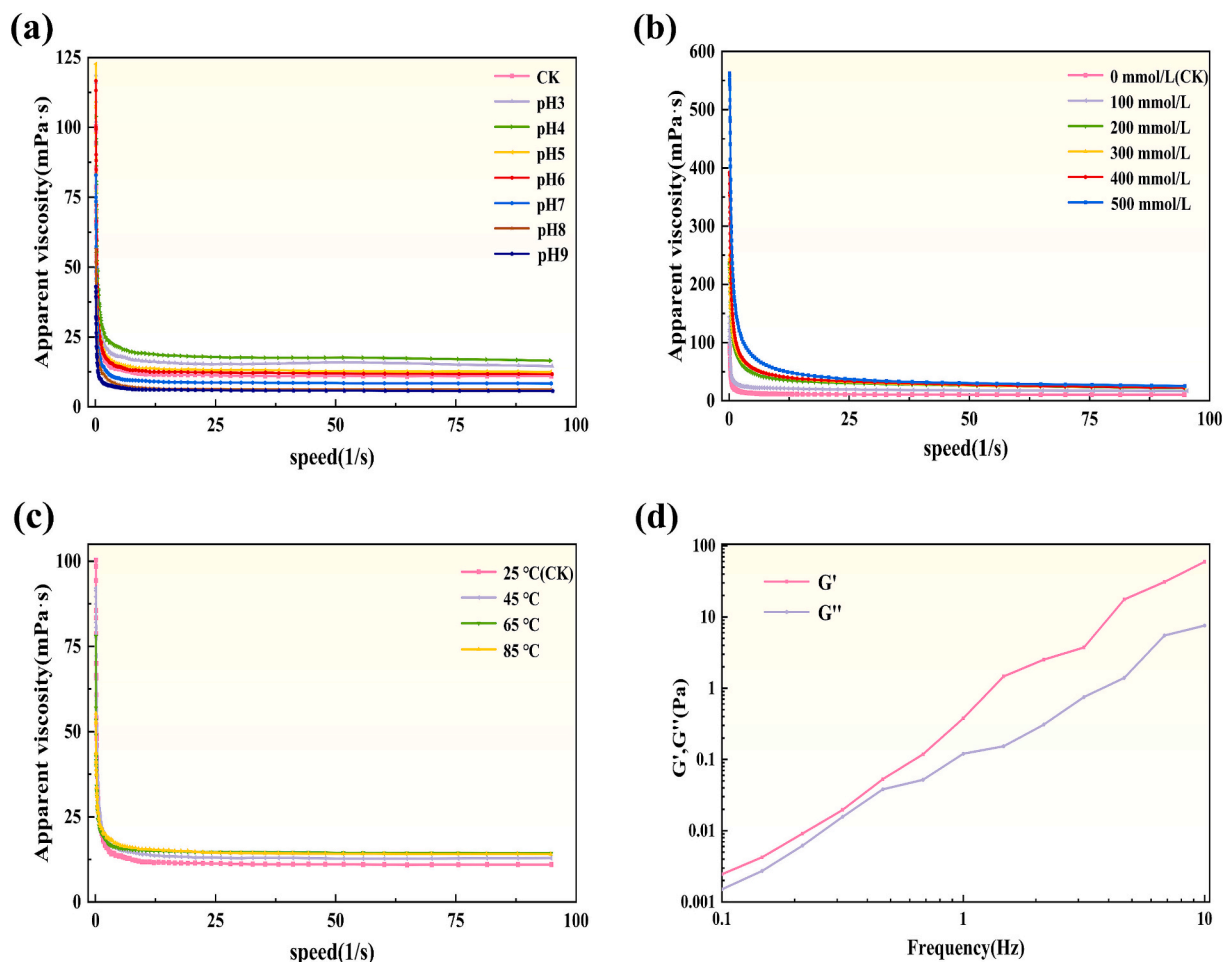


Fig. 8. Effect of pH value (a), NaCl concentration (b) and temperature (c) on the apparent viscosity of nanoemulsions. Frequency dependence of the storage modulus (G') and loss modulus (G'') of the nanoemulsion (d).

Furthermore, pH exerted a highly significant influence on both the viscosity coefficient K and the degree of shear thinning n of the nanoemulsion.

Fig. 8b displays the effect of NaCl concentration on the apparent viscosity of the nanoemulsion. As shown in the figure, at a shear rate of 0.1 s^{-1} , the apparent viscosity of the emulsion increases with increasing NaCl concentration. The viscosity exhibits good parallelism with the control group at an NaCl concentration of 100 mmol/L. Above this concentration, the emulsion viscosity increases significantly with increasing salt concentration. This behavior may be related to the reduction in interphase surface tension across the oil-water interface caused by increased salinity. The increase in nanoemulsion viscosity with salinity is also associated with charge shielding effects (Onaizi, 2022). Due to the neutralization of Na^+ with the negative charge of the droplets, the electrostatic repulsion between nanoemulsion droplets decreases, causing some oil droplets to flocculate and form droplet aggregates. Through flocculation, some water molecules become trapped within the resulting flocs. Consequently, the “effective hydrodynamic volume fraction” of the emulsion droplets increases, leading to higher emulsion viscosity (Al-Yaari et al., 2015).

Relevant data were analyzed using the power-law model. As shown in Table 3, the nanoemulsion exhibited a flow behavior index n below 1 at all tested NaCl concentrations, indicating it consistently behaved as a pseudoplastic fluid (shear-thinning). However, the increase in salt concentration significantly affected the internal structure of the nanoemulsion, manifested by a sharp rise in the viscosity coefficient K . This signifies that the emulsion system's viscosity substantially increased with rising salt concentration.

Fig. 8c shows that at a shear rate of 0.1 s^{-1} , the apparent viscosity decreases with increasing heat treatment temperature. As temperature rises, the viscosity of the nanoemulsion decreases, which is related to increased Brownian motion of the droplets. The thermal energy of the nanoemulsion increases with rising temperature, thereby enhancing the migration rate of the nanoemulsion droplets and breaking down the emulsion flocs. This enhanced fluidity directly results in reduced viscosity (Onaizi, 2022).

Relevant data were analyzed using the power-law model. As shown in Table 4, This nanoemulsion exhibits pseudoplastic behavior ($n < 1$) at all tested temperatures, demonstrating shear thinning characteristics. Temperature variations primarily affect the internal structural strength of the fluid, with relatively minor impacts on overall viscosity. As temperature increases, the emulsion's flow behavior progressively approaches that of a Newtonian fluid.

To further characterize the mechanical behavior of the nanoemulsion, oscillatory rheological measurements were performed by frequency sweep analysis. As shown in Fig. 8d, both the storage modulus (G') and loss modulus (G'') increased with increasing frequency, indicating a frequency-dependent viscoelastic response of the nanoemulsion. Notably, G' remained consistently higher than G'' throughout the tested frequency range, suggesting that the system exhibited elastic-dominant viscoelastic behavior with weak gel-like characteristics. This result implies the presence of a relatively stable internal structure within the nanoemulsion, which may help limit droplet mobility and reduce the tendency toward aggregation or phase separation. Therefore, the oscillatory rheological results further support the good physical stability of

Table 3

Herschel-Bulkley model parameters of nanoemulsions with different NaCl concentration.

NaCl concentration (mmol/L)	$K \text{ (Pa}\cdot\text{s}^n)$	n	R^2
0 (CK)	25.33 ± 0.79	0.4364 ± 0.0186	0.9474
100	40.19 ± 1.31	0.5204 ± 0.0199	0.9073
200	99.27 ± 0.75	0.5868 ± 0.0047	0.9931
300	104.92 ± 0.29	0.6202 ± 0.0017	0.9989
400	128.04 ± 1.02	0.4895 ± 0.0048	0.9958
500	196.82 ± 2.70	0.4841 ± 0.0082	0.9887

Table 4

Herschel-Bulkley model parameters of nanoemulsions with different temperature.

Temperature ($^{\circ}\text{C}$)	$K \text{ (Pa}\cdot\text{s}^n)$	n	R^2
25 (CK)	25.4154 ± 0.8604	0.4391 ± 0.0201	0.9371
45	30.6412 ± 0.7446	0.5254 ± 0.0148	0.9478
65	25.5183 ± 0.6190	0.7322 ± 0.0149	0.7984
85	25.8275 ± 0.3744	0.7674 ± 0.0087	0.8979

the carotenoid-loaded nanoemulsion during processing and storage.

3.11. *In vitro* gastrointestinal digestion

As shown in Fig. 7f, the carotenoid release from the nanoemulsion exhibited a clear digestion-stage-dependent behavior during simulated gastrointestinal digestion. In the oral phase, carotenoid release remained at a relatively low level and increased only slightly with digestion time, indicating that the nanoemulsion was able to protect carotenoids effectively under mild conditions. In the gastric phase, the release rate showed a modest increase, but the overall carotenoid release was still limited, suggesting that the acidic environment and pepsin-rich conditions had only a minor effect on the structural integrity of the nanoemulsion. By contrast, during the intestinal phase, carotenoid release increased sharply and reached the highest level among the three digestion stages. This pronounced increase may be attributed to the action of bile salts and pancreatic enzymes, which can promote the disintegration of the nanoemulsion structure and facilitate the liberation of encapsulated carotenoids. Similar behavior has been reported in previous studies, where the intestinal environment was identified as the main site for carotenoid release and potential absorption (Von Lintig et al., 2019).

Overall, these results indicate that the nanoemulsion provided effective protection for carotenoids during the oral and gastric phases, while enabling substantial release in the intestinal phase. This digestion-responsive release behavior is favorable for functional food applications, as it may enhance carotenoid bioaccessibility and improve the likelihood of intestinal absorption.

4. Conclusions

This study employs a novel green solvent, DES, to extract carotenoid compounds from orange peel. The extraction yield is determined using ultraviolet spectrophotometry, and extraction conditions are optimized. Concurrently, the extracted solution undergoes antioxidant and stability testing. Additionally, solid recovery of carotenoid compounds extracted from orange peel was performed, and the resulting carotenoid extract was used as the oil phase to prepare carotenoid-loaded nanoemulsions. The selected DES in this study was decanoic acid: lauric acid = 2:1. The optimized extraction conditions were: extraction temperature $40.50 \text{ }^{\circ}\text{C}$, liquid-to-solid ratio 10.5:1 mL/g, and extraction time 16.80 min. Under these optimal conditions, the extraction yield reached $60.98 \pm 0.87 \text{ mg/100 g}$. Using a high-energy emulsification method, nanoemulsions were prepared with carotenoid-rich DES as the oil phase and Tween 80 as the surfactant. The effect of Tween 80 addition on the emulsion was investigated. Dyeing experiments confirmed the nanoemulsion was O/W type; with an average particle size of $76.78 \pm 1.58 \text{ nm}$, PDI value of 0.15 ± 0.01 , and zeta potential of $-42.56 \pm 1.59 \text{ mV}$. The system exhibited stability and uniform distribution, meeting nanoemulsion requirements. TEM revealed spherical or ellipsoidal droplets with good formation. Additionally, this study investigated the effects of pH, temperature, and NaCl concentration on the stability of nanoemulsions. Results indicate relative stability at pH 3–7; unstable aggregation occurs in high-salt environments ($>200 \text{ mmol/L}$); and relative stability is maintained under certain heating conditions. Finally, analysis of the rheological properties of the nanoemulsion revealed that its viscosity decreases with

increasing shear rate, exhibiting excellent shear thinning behavior. This characteristic enhances the applicability of the nanoemulsion in fields such as food and pharmaceuticals. Additionally, the rheological properties provide further evidence of the stability of the nanoemulsion under specific conditions. Overall, this study provides new insights into the efficient extraction, recovery, and utilization of carotenoids and offers a basis for the development of carotenoid delivery systems in the food industry. Notably, the pH-switchable nature of the solvent system facilitates downstream carotenoid recovery and supports a more integrated extraction-recovery process, thereby showing potential advantages over conventional methods that often involve additional solvent replacement or more cumbersome separation procedures. However, the environmental and techno-economic performance of the proposed process was not systematically assessed in the present study. Future work should further compare this system with conventional organic solvent extraction in terms of energy consumption, carbon footprint, toxicity, and process cost to better evaluate its sustainability and industrial feasibility.

CRedit authorship contribution statement

Yanhong Liu: Project administration, Funding acquisition, Conceptualization. **Yixin Wang:** Writing – review & editing, Writing – original draft, Investigation, Formal analysis. **Ruonan Wu:** Validation, Investigation. **Hongyu Qian:** Software, Data curation. **Mengyao Liu:** Resources, Methodology. **Lingling Zhang:** Resources, Methodology. **Shuo Wang:** Validation, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.foodchem.2026.149814>.

Data availability

Data will be made available on request.

References

- Abbott, A. P., Boothby, D., Capper, G., Davies, D. L., & Rasheed, R. K. J. (2004). Deep eutectic solvents formed between choline chloride and carboxylic acids: Versatile alternatives to ionic liquids. *Journal of the American Chemical Society*, 126(29), 9142–9147. <https://doi.org/10.1021/ja048266j>
- Al-Yaari, M., Hussein, I. A., Al-Sarkhi, A., Abbad, M., & Chang, F. (2015). Effect of water salinity on surfactant-stabilized water–oil emulsions flow characteristics. *Experimental Thermal and Fluid Science*, 64, 54–61. <https://doi.org/10.1016/j.expthermflusci.2015.02.001>
- Ariffin, T. S. T., Yahya, E., & Husin, H. (2016). The rheology of light crude oil and water-in-oil-emulsion. *Procedia Engineering*, 148, 1149–1155. <https://doi.org/10.1016/j.proeng.2016.06.614>
- Bai, L., Geng, S., Zhou, Y., Ma, H., & Liu, B. (2024). Ultrasound-assisted fabrication and stability evaluation of okra seed protein stabilized nanoemulsion. *Ultrasonics Sonochemistry*, 104, Article 106807. <https://doi.org/10.1016/j.ultsonch.2024.106807>
- Boonlao, N., Ruktanonchai, U. R., & Anal, A. K. (2022). Enhancing bioaccessibility and bioavailability of carotenoids using emulsion-based delivery systems. *Colloids and Surfaces B: Biointerfaces*, 209, Article 112211. <https://doi.org/10.1016/j.colsurfb.2021.112211>
- Borthakur, P., Boruah, P. K., Sharma, B., & Das, M. R. (2016). *Nanoemulsion: Preparation and its application in food industry* (pp. 153–191). Academic Press. <https://doi.org/10.1016/B978-0-12-804306-6.00005-2>
- Bu, G., Zhao, C., Wang, M., Yu, Z., Yang, H., & Zhu, T. (2023). The development and properties of nanoemulsions stabilized with glycated soybean protein for carrying β -carotene. *Journal of Food Engineering*, 345, Article 111411. <https://doi.org/10.1016/j.jfoodeng.2023.111411>
- Chang, Y., McLandsborough, L., & McClements, D. J. (2013). Physicochemical properties and antimicrobial efficacy of carvacrol nanoemulsions formed by spontaneous emulsification. *Journal of Agricultural and Food Chemistry*, 61(37), 8906–8913. <https://doi.org/10.1021/jf402147p>
- Chen, X., Wang, R., & Tan, Z. (2023). Extraction and purification of grape seed polysaccharides using pH-switchable deep eutectic solvents-based three-phase partitioning. *Food Chemistry*, 412, Article 135557. <https://doi.org/10.1016/j.foodchem.2023.135557>
- Cvjetko Bubalo, M., Vidović, S., Radojčić Redovniković, I., & Jokić, S. (2018). New perspective in extraction of plant biologically active compounds by green solvents. *Food and Bioprocess Processing*, 109, 52–73. <https://doi.org/10.1016/j.fbp.2018.03.001>
- Dai, Y., Witkamp, G.-J., Verpoorte, R., & Choi, Y. H. (2015). Tailoring properties of natural deep eutectic solvents with water to facilitate their applications. *Food Chemistry*, 187(15), 14–19. <https://doi.org/10.1016/j.foodchem.2015.03.123>
- Desmarchelier, C., & Borel, P. (2017). Overview of carotenoid bioavailability determinants: From dietary factors to host genetic variations. *Trends in Food Science & Technology*, 69, 270–280. <https://doi.org/10.1016/j.tifs.2017.03.002>
- Foudazi, R., Qavi, S., Masalova, I., & Malkin, A. Y. (2015). Physical chemistry of highly concentrated emulsions. *Advances in Colloid and Interface Science*, 220, 78–91. <https://doi.org/10.1016/j.cis.2015.03.002>
- Gao, M.-Z., Cui, Q., Wang, L.-T., Meng, Y., Yu, L., Li, Y.-Y., & Fu, Y.-J. (2020). A green and integrated strategy for enhanced phenolic compounds extraction from mulberry (*Morus alba* L.) leaves by deep eutectic solvent. *Microchemical Journal*, 154, Article 104598. <https://doi.org/10.1016/j.microc.2020.104598>
- Guo, Q., Bayram, I., Shu, X., Su, J., Liao, W., Wang, Y., & Gao, Y. (2022). Improvement of stability and bioaccessibility of β -carotene by curcumin in pea protein isolate-based complexes-stabilized emulsions: Effect of protein complexation by pectin and small molecular surfactants. *Food Chemistry*, 367, Article 130726. <https://doi.org/10.1016/j.foodchem.2021.130726>
- Hu, Y., Chen, X., & Tan, Z. (2023). Three-phase partitioning constructed by pH-responsive deep eutectic solvents and sugars for purification of radish (*Raphanus sativus* L.) peroxidase. *Separation and Purification Technology*, 322, Article 124353. <https://doi.org/10.1016/j.seppur.2023.124353>
- Huang, H., Guo, S., Xu, Y., Ettoumi, F.-E., Fang, J., Yan, X., Xie, Z., Luo, Z., & Cheng, K. (2024). Valorization and protection of anthocyanins from strawberries (*Fragaria × ananassa* Duch.) by acidified natural deep eutectic solvent based on intermolecular interaction. *Food Chemistry*, 447, Article 138971. <https://doi.org/10.1016/j.foodchem.2024.138971>
- Huang, H., Wang, Y., Chen, Y., Zhou, T., Wang, X., Zhou, J., Pan, J., Luo, Z., & Cheng, K. (2025). Incremental extraction of flavonoids from *Penthorum chinense* Pursh for alleviating dextran sulfate sodium-induced mouse colitis by regulating gut microbiota. *Food Bioscience*, 64, Article 105897. <https://doi.org/10.1016/j.fbio.2025.105897>
- Huang, H., Ying, P., Wang, Y., Wu, Q., Wang, L., & Fu, X. (2025). Temperature-dependent convection induced incremental extraction of anthocyanins from *Melastoma dodecandrum* Lour. Based on recyclable natural deep eutectic system. *Food Chemistry*, 484, Article 144331. <https://doi.org/10.1016/j.foodchem.2025.144331>
- Huang, H., Zhu, Y., Fu, X., Zou, Y., Li, Q., & Luo, Z. (2022). Integrated natural deep eutectic solvent and pulse-ultrasonication for efficient extraction of crocins from gardenia fruits (*Gardenia jasminoides* Ellis) and its bioactivities. *Food Chemistry*, 380, Article 132216. <https://doi.org/10.1016/j.foodchem.2022.132216>
- Jafari, S. M., & McClements, D. J. (2017). *Chapter one - Nanotechnology approaches for increasing nutrient bioavailability*. 81 pp. 1–30). Academic Press. <https://doi.org/10.1016/bs.afnr.2016.12.008>
- Jessop, P. G., Mercer, S. M., & Heldebrandt, D. J. (2012). CO₂-triggered switchable solvents, surfactants, and other materials. *Energy & Environmental Science*, 5(6), 7240–7253. <https://doi.org/10.1039/c2ee03442a>
- Li, D., Zhao, Y., Wang, X., Tang, H., Wu, N., Wu, F., Yu, D., & Elfalleh, W. (2020). Effects of (+)-catechin on a rice bran protein oil-in-water emulsion: Droplet size, zeta-potential, emulsifying properties, and rheological behavior. *Food Hydrocolloids*, 98, Article 105306. <https://doi.org/10.1016/j.foodhyd.2019.105306>
- Li, X.-M., Li, X., Wu, Z., Wang, Y., Cheng, J.-S., Wang, T., & Zhang, B. (2020). Chitosan hydrochloride/carboxymethyl starch complex nanogels stabilized Pickering emulsions for oral delivery of β -carotene: Protection effect and in vitro digestion study. *Food Chemistry*, 315(15), Article 126288. <https://doi.org/10.1016/j.foodchem.2020.126288>
- Li, Y., Hu, K., Huang, C., Hu, Y., Ji, H., Liu, S., & Gao, J. (2022). Improvement of solubility, stability and antioxidant activity of carotenoids using deep eutectic solvent-based microemulsions. *Colloids and Surfaces B: Biointerfaces*, 217, Article 112591. <https://doi.org/10.1016/j.colsurfb.2022.112591>
- Li, Z., Dai, L., Wang, D., Mao, L., Gao, J., & Yanxiang. (2018). Stabilization and rheology of concentrated emulsions using the natural emulsifiers Quillaja saponins and rhamnolipids. *Journal of Agricultural and Food Chemistry*, 66(15), 3922–3929. <https://doi.org/10.1021/acs.jafc.7b05291>
- Longeras, O., Gautier, A., Ballerat-Busserolles, K., & Andanson, J.-M. (2020). Deep eutectic solvent with Thermo-switchable hydrophobicity. *ACS Sustainable Chemistry*

- & Engineering, 8(33), 12516–12520. <https://doi.org/10.1021/acssuschemeng.0c03478>
- Ma, Y., Wang, Q., & Zhu, T. (2019). Comparison of hydrophilic and hydrophobic deep eutectic solvents for pretreatment determination of sulfonamides from aqueous environments. *Analytical Methods*, 11(46), 5901–5909. <https://doi.org/10.1039/c9ay02244a>
- Mazzarino, L., da Silva Pitz, H., Lorenzen Voytena, A. P., Dias Trevisan, A. C., Ribeiro-Do-Valle, R. M., & Maraschin, M. (2018). Jaboticaba (*Plinia peruviana*) extract nanoemulsions: Development, stability, and in vitro antioxidant activity. *Drug Development and Industrial Pharmacy*, 44(4), 643–651. <https://doi.org/10.1080/03639045.2017.1405976>
- Md Yusoff, M. H., Gan, C.-Y., & Shafie, M. H. (2023). Characterization of citric acid monohydrate-glycerol based deep eutectic solvents which could be used as an extraction medium for hydrophilic bioactive components. *Journal of Molecular Liquids*, 389, Article 122879. <https://doi.org/10.1016/j.molliq.2023.122879>
- Mudalige, T., Qu, H., Van Haute, D., Ansar, S. M., Paredes, A., & Ingle, T. (2019). Nanomaterials for food applications. In *Characterization of nanomaterials: Tools and challenges* (pp. 313–353). <https://doi.org/10.1016/B978-0-12-814130-4.00011-7>
- Murador, D. C., Salafia, F., Zoccali, M., Martins, P. L. G., Ferreira, A. G., Dugo, P., ... Giuffrida, D. (2019). Green extraction approaches for carotenoids and esters: Characterization of native Composition from Orange Peel. *Antioxidants*, 8(12), 613. <https://doi.org/10.3390/antiox8120613>
- Nejadmansouri, M., Hosseini, S. M. H., Niakosari, M., Yousefi, G. H., & Golmakani, M. T. (2016). Physicochemical properties and oxidative stability of fish oil nanoemulsions as affected by hydrophilic lipophilic balance, surfactant to oil ratio and storage temperature. *Colloids and Surfaces A: Physicochemical and Engineering Aspects*, 506, 821–832. <https://doi.org/10.1016/j.colsurfa.2016.07.075>
- Onaizi, S. A. (2022). Effect of salinity on the characteristics, pH-triggered demulsification and rheology of crude oil/water nanoemulsions. *Separation and Purification Technology*, 281, Article 119956. <https://doi.org/10.1016/j.seppur.2021.119956>
- Ordóñez-Lozada, M. I., Bobrowski Rodrigues, D., Maldonado, I. R., Fernandes, T. A., de Alencar, E. R., & de Oliveira, L. L. (2025). Stability of lutein-based nanoemulsions under thermal and UVC-light exposure: Effect of lutein esterification and oil phase constitution. *Food Chemistry*, 492, Article 145401. <https://doi.org/10.1016/j.foodchem.2025.145401>
- Paiva, A., Craveiro, R., Aroso, I., Martins, M., Reis, R. L., & Duarte, A. R. C. (2014). Natural deep eutectic solvents – Solvents for the 21st century. *ACS Sustainable Chemistry & Engineering*, 2(5), 1063–1071. <https://doi.org/10.1021/sc500096j>
- Sachindra, N. M., & Mahendrakar, N. S. (2010). Stability of carotenoids recovered from shrimp waste and their use as colorant in fish sausage. *Journal of Food Science and Technology*, 47(1), 77–83. <https://doi.org/10.1007/s13197-010-0019-z>
- Sadeghian, S. F., Majdinasab, M., Nejadmansouri, M., & Hosseini, S. M. H. (2023). Effects of natural antioxidants and high-energy fabrication methods on physical properties and oxidative stability of flaxseed oil-in-water nanoemulsions. *Ultrasonics Sonochemistry*, 92, Article 106277. <https://doi.org/10.1016/j.ultsonch.2022.106277>
- Santana, A. P. R., Mora-Vargas, J. A., Guimarães, T. G. S., Amaral, C. D. B., Oliveira, A., & Gonzalez, M. H. (2019). Sustainable synthesis of natural deep eutectic solvents (NADES) by different methods. *Journal of Molecular Liquids*, 293, Article 111452. <https://doi.org/10.1016/j.molliq.2019.111452>
- Shanmugam, A., & Ashokkumar, M. (2014). Ultrasonic preparation of stable flax seed oil emulsions in dairy systems – Physicochemical characterization. *Food Hydrocolloids*, 39, 151–162. <https://doi.org/10.1016/j.foodhyd.2014.01.006>
- Sharma, N., Kaur, G., & Khatkar, S. K. (2021). Optimization of emulsification conditions for designing ultrasound assisted curcumin loaded nanoemulsion: Characterization, antioxidant assay and release kinetics. *LWT*, 141, Article 110962. <https://doi.org/10.1016/j.lwt.2021.110962>
- Stupar, A., Šeregelj, V., Ribeiro, B. D., Pezo, L., Cvetanović, A., Mišan, A., & Marrucho, I. (2021). Recovery of β -carotene from pumpkin using switchable natural deep eutectic solvents. *Ultrasonics Sonochemistry*, 76, Article 105638. <https://doi.org/10.1016/j.ultsonch.2021.105638>
- Su, S., Lu, J., Yin, X., Wang, M., Wang, F., & Pang, Y. (2025). Encapsulation of β -carotene in chitosan Pickering nanoemulsion: Effect of process parameters and their stability. *Food Chemistry*, 500, Article 147505. <https://doi.org/10.1016/j.foodchem.2025.147505>
- Van Osch, D. J. G. P., Zubeir, L. F., van den Bruinhorst, A., Rocha, M. A. A., & Kroon, M. C. (2015). Hydrophobic deep eutectic solvents as water-immiscible. *Green Chemistry*, 17(9), 4518–4521. <https://doi.org/10.1039/c5gc01451d>
- Vilas-Boas, A. A., Magalhães, D., Gómez-García, R., Campos, D. A., Correia, M., & Pintado, M. (2025). Upcycled orange peel ingredients: A scoping review on phytochemical composition, extraction techniques, and biorefinery strategies. *Foods*, 14(21), 3766. <https://doi.org/10.3390/foods14213766>
- Víñas-Ospino, A., Panić, M., Radojčić-Redovniković, I., Blesa, J., & Esteve, M. J. (2023). Using novel hydrophobic deep eutectic solvents to improve a sustainable carotenoid extraction from orange peels. *Food Bioscience*, 53, Article 102570. <https://doi.org/10.1016/j.fbio.2023.102570>
- Von Lintig, J., Moon, J., Lee, J., & Ramkumar, S. (2019). Carotenoid metabolism at the intestinal barrier. *Biochimica et Biophysica Acta (BBA)-Molecular and Cell Biology of Lipids*, 1861(11), Article 158580. <https://doi.org/10.1016/j.bbalip.2019.158580>
- Wang, P., Wang, J., Hu, H., Ding, N., Wu, F., Fang, G., Mao, J., Wu, C., & Liu, X. (2026). Encapsulation of quercetin in zein/sodium caseinate nanoparticles: Fabrication, characterization, stability and controlled release properties. *Food Research International*, 230, Article 118608. <https://doi.org/10.1016/j.foodres.2026.118608>
- Wu, H., Zhang, X., Cai, Z., Tang, J., Che, Z., Liu, P., Wu, T., & Shao, W. (2025). Preparation and characterization of flaxseed oil emulsions stabilized by chitosan/gelatin complexes and evaluation of oxidation stability. *International Journal of Biological Macromolecules*, 309, Article 142918. <https://doi.org/10.1016/j.ijbiomac.2025.142918>
- Xu, M., Ran, L., Chen, N., Fan, X., Ren, D., & Yi, L. (2019). Polarity-dependent extraction of flavonoids from citrus peel waste using a tailor-made deep eutectic solvent. *Food Chemistry*, 297(1), Article 124970. <https://doi.org/10.1016/j.foodchem.2019.124970>
- Yang, T., Liu, C., Zheng, Y., Liu, T. C., Li, K., Liu, J., ... Zhou, P. (2023). Effect of WPI/Tween 80 mixed emulsifiers on physicochemical stability of ginsenosides nanoemulsions. *Food Bioscience*, 53, Article 102519. <https://doi.org/10.1016/j.fbio.2023.102519>
- Zeng, C., Liu, Y., Ding, Z., Xia, H., & Guo, S. (2021). Physicochemical properties and antibacterial activity of hydrophobic deep eutectic solvent-in-water nanoemulsion. *Journal of Molecular Liquids*, 338, Article 116950. <https://doi.org/10.1016/j.molliq.2021.116950>
- Zhang, H., Zhao, W., Liu, L., Wen, W., Jing, X., & Wang, X. (2023). Switchable deep eutectic solvents for sustainable extraction of β -carotene from millet. *Microchemical Journal*, 187, Article 108369. <https://doi.org/10.1016/j.microc.2022.108369>
- Zhao, D., Ge, Y., Xiang, X., Dong, H., Qin, W., & Zhang, Q. (2022). Structure and stability characterization of pea protein isolate-xylan conjugate-stabilized nanoemulsions prepared using ultrasound homogenization. *Ultrasonics Sonochemistry*, 90, Article 106195. <https://doi.org/10.1016/j.ultsonch.2022.106195>
- Zhou, Y., Bai, L., Geng, S., & Liu, B. (2025). Interaction of camellianin A and lysozyme: Binding mechanism and its application in nanoemulsions. *Food Chemistry*, 475(30), Article 143265. <https://doi.org/10.1016/j.foodchem.2025.143265>
- Ziani, K., Chang, Y., McLandsborough, L., & McClements, D. J. (2011). Influence of surfactant charge on antimicrobial efficacy of surfactant-stabilized thyme oil nanoemulsions. *Journal of Agricultural and Food Chemistry*, 59(11), 6247–6255. <https://doi.org/10.1021/jf200450m>